BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Performance

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is FTP used for?

2. How does phishing trick people?
   A) Tries every possible password
   B) Floods a server with traffic
   C) Repairs damaged files
   D) Pretends to be trusted

3. "Bluetooth is normally the best choice for a high-capacity link between distant school sites." Correct this. Explain why.

4. A network shows 2001:db8::7. "IPv6 uses four decimal numbers separated by dots." Correct this. How is this address written?

Today's objectives

Explain the core ideas of network performance using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Bandwidth
The available data transfer capacity of a connection over time.

Latency
Delay before data begins or a response is received.

Wired and wireless
Wired links tend to be less affected by wireless interference; wireless provides mobility.

Performance factors
Traffic, distance, medium, devices, interference and service contention affect observed speed.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Bandwidth
The available data transfer capacity of a connection over time.

Latency
Delay before data begins or a response is received.

Wired and wireless
Wired links tend to be less affected by wireless interference; wireless provides mobility.

Performance factors
Traffic, distance, medium, devices, interference and service contention affect observed speed.

I Do - Worked model

A 24 Mb file travels at a constant 8 Mb/s. Calculate ideal transfer time.

- Identify units: both are megabits.
- Use time = file size / transfer rate.
- Compute 24 / 8 = 3 seconds.
- Explain this ideal figure excludes latency/overhead.

We Do - Guided practice

1. A 60 Mb file travels at 12 Mb/s. Calculate ideal time.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the calculation of transfer time in seconds.

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb * speedMbps
print(seconds)
```

You Do - Independent exam questions

Q2. Explain the difference between bandwidth and latency. [2 marks]

Q3. Calculate time for a 40 Mb file over 10 Mb/s ideal link. [2 marks]

Q4. Give two factors reducing Wi-Fi performance. [2 marks]

Exit Ticket - True or False?

Bandwidth and latency are exactly the same measurement.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Transferring files between networked computers.

2. D - Pretends to be trusted

3. Bluetooth suits short-range links, such as keyboards or headphones. Check range, data rate and power needs.

4. IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN.

I Do - Explain the model

Identify units: both are megabits.

Use time = file size / transfer rate.

Compute 24 / 8 = 3 seconds.

Explain this ideal figure excludes latency/overhead.

We Do - Model answers

1. 5 seconds.

Code First - Answer

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb / speedMbps
print(seconds)
```

Expected output: 3

You Do - Self-mark

Q2: Bandwidth is data capacity per unit time; latency is delay before response/transfer.

Q3: 40 / 10 = 4 seconds.

Q4: Interference, weak signal/distance, number of users or obstacles.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.